

Story

Two friends arrive in a new village with nothing. To build a house, they must gather materials by exploring the village and speaking with the locals. Each villager has a unique job related to a specific building material. There are four mini-games: one to collect wood, one to generate electricity, one to obtain ink for painting, and another to gather items for the garden. After completing all the mini-games, the friends finally get their house.

Game Interface

Initial Menu: The first screen presented to the player with two options:

- Start: To begin the game
- Quit: To exit the game

Island (Main World): A 3D open-world island where characters can walk and jump to explore the environment. Materials appear in a HUD at the bottom of the screen. Players interact with villagers to receive guidance and access mini-games. There's also a button to return to the main menu at any time.

Electrical Connections (Minigame): A 2.5D minigame where each player controls a tile selector that allows the players to rotate the tiles and route the energy. The selectors have different colors for easy differentiation and identification of their location. There is also a timer in the top left of the screen indicating the time the players have left to solve the puzzle. Each tile visually changes its rotation once the players interact with it. If a tile is powered, its line is yellow, it is black, and colors change if the power is not given to the tile, either by being too far from a source or not being connected to a powered tile. All generators have a yellow circle indicating they are a source of power, and the house is initially black and turns yellow if the puzzle is completed.

Timber Time (Minigame): A 2D minigame where each player has an axe, and each log that needs to be cut has assigned cuts to both players. These are indicated by different color pointers, which disappear when the cut is made in the correct place. Cutting the log also results in it becoming visually cut. The UI has the score on the top left-hand side of the screen and the time left in the top center.

Painting Walls (Minigame): A 2.5D cooperative game where both players aim to match a color pattern on a wall within a time limit. Each player controls a character that can move around the scene, collect colors from buckets, and apply them to wall tiles. A visual pattern displayed on-screen guides the players. Each tile gives immediate feedback when painted. A timer is displayed at the side, and color indicators are shown on the UI for each player. The scene also includes sound effects for painting and a background track to maintain immersion. The game ends once all tiles are correctly painted.

ZenGarden (Minigame): A 2.5D cooperative sokoban-style puzzle where players must push pots onto their respective color-coded targets. The game interface includes a top-down view, a relaxing background track, and soft visuals that promote focus. Players receive audio-visual feedback when pots are placed correctly, and a HUD shows remaining time and optional retry/forfeit buttons. When all pots are correctly positioned, the game transitions automatically to the result screen.

Minigame End Screen: Each minigame has an end screen, and they all follow a pattern. The user is presented with a text indicating the result of the minigame and, in some cases, the score obtained in said minigame. The user also has a button at their disposal to return to the Island. If the user does not interact with anything on the page, they will be taken to the Island either way after a set amount of time, clearly displayed at the bottom left of the screen.

Final Cutscene: Once all materials are collected, a short cutscene shows the house being built. This is the only non-interactive part of the game, designed to reward the player with a satisfying moment.

Final Menu: After the cutscene, the player is presented with two options:

- Restart: Resets all progress and collected materials
- Quit: Exits the game

Character Abilities and Items

Characters can walk and jump around the island.

They can collect different types of materials:

- Wood
- Electricity
- Paint
- Gardening items

In the Electrical Connections minigame, the player controls the tile selector by moving it on the board and is able to interact with it to rotate the tiles 90° to the right. Each player can only move the selector on their side of the board. At the end of the minigame, the player acquires the previously mentioned material, Electricity.

In the Timber Time minigame, each player controls an axe, which they can use to cut the log. They can precisely move this axe to cut in the correct place. They cannot, however, move over to the other side of the screen, which is reserved for the other player - each player must cover the marks on their side.

The painting walls minigame is a cooperative minigame where players must reproduce a color pattern on a wall using brushes and paint. Each player can move independently and interact with tiles on the wall to either pick a color from a paint source or apply it to a tile. Players must match the correct color on each tile based on a pattern displayed on screen. The game encourages coordination as players work together to complete the wall. Feedback is given visually and via a success screen once the pattern is correctly completed. The wall has a slightly worn-down aesthetic, reinforcing the idea of restoring or decorating part of the village.

The ZenGarden minigame is a 2.5D grid-based cooperative puzzle where both players navigate on a top-down layout of walkable tiles. Each level contains colored targets embedded in the ground and movable pots scattered around walls and obstacles. Players use their character to push pots one tile at a time onto matching targets; walls and other pots block movement, and pushing a pot into water destroys it. Player 1 moves with WASD,

Player 2 with the arrow keys. Once all pots rest on their correct targets, the minigame ends immediately and awards the gardening items.

At the end, when the players complete all the minigames, they get the house.

Game World description and design

The game offers a calm and relaxing experience. In the main island, players hear ambient bird sounds and soft background music. Each mini-game is also accompanied by peaceful music to keep the experience soothing, even when the player faces challenges.

The 3D island encourages exploration, and players must find and speak with villagers to unlock each mini-game. A HUD displays all materials the player has collected so far.

The minigames themselves seek to differentiate one from another by providing different experiences along with a world design that is adapted to the purpose/objective of the minigame.

The final scene, a non-interactive cutscene, lets players appreciate the house they helped build, providing a sense of closure and reward.

To help players understand the mechanics of each mini-game without needing to memorize the controls, a short tutorial is shown every time a mini-game begins. This tutorial briefly explains the rules and how to play. If the player prefers to skip it, a skip button is available.

Game Level Design (for each level)

Electrical connections is a cooperative logic-based minigame that makes two players connect the energy using the pipes. Each player can control a tile selector and rotate the tile 90° at a time, which is accompanied by a rotating tile sound in order to give the player some audio feedback. Each player can only control half of the board, creating the necessity to cooperate to win the game. There are multiple generators scattered throughout the map to confuse players, as not all of them need to be connected to win. Powering a tile changes its color to yellow to inform the player that it is powered. The power wears off after some tiles, so connecting more than one generator is necessary, and connecting to a generator for the first time will produce a sound to give audio feedback. To add some difficulty, a timer was added to make players get better immersed in the game, and once the timer enters the last 10 seconds, a sound starts playing, warning the player that the game is about to end. In order to win, the players must connect all tiles, leaving no ends open and having all of them powered, including the house. Background music is also playing, maintaining the relaxing experience intended for the game. Regarding visual design, a brick wall with some vegetation was chosen in order to give the impression that this circuit has been disconnected for a long time, and nature is starting to take control of its space. The player characters were also added to the scene, allowing for continuity of the previous scene on the main island. The design of the generators was kept simple so as not to overly clutter the scene. The game was chosen to be in 2.5D to increase the immersion of the player in the minigame.

Timber Time is a cooperative minigame where the players need to race against the (60-second) clock to cut as many logs as possible to beat the minimum required 400 points. Each player controls an axe they can use to cut, and can control it to cut precisely along the log, though they cannot move over to the other player's side. This cut is accompanied by a cutting sound effect, as well as the log becoming visually cut. Each log has several marks on it where the players need to cut to complete it (1-3 on each side), and if completed gives 50 points (accompanied by a success sound effect). However, if the players cut it in the wrong spot, it is lost, and they get a 10-point penalty as well as a failure sound effect. When they finish cutting the log or cut incorrectly, the platform holding the log opens, and the log falls. As it closes back up, a new log appears on it, with new spots marked for cuts. When the time is at 3 seconds left, a sound plays to warn the players their time is running out, and when it ends, it skips to the game over screen. To create immersion, the background of the minigame is a forest, and background music is playing at all times. The decision to make the game 2D as opposed to 2.5D was meant to facilitate the finding of assets suitable for the minigame (as this had been the main issue with the previous version of this minigame, which we had to cut out and replace).

Painting Walls is a cooperative minigame where players work together to paint a wall by matching a predefined color pattern in a very short time (2:30 minutes), making it quite challenging. It was made in 2.5D. Each tile on the wall has a target color, and players must pick up the correct paint from sources and apply it to the corresponding tiles. Each player controls a character that can move around the scene, interact with buckets to collect colors, and paint the tiles. Each player only has access to two color buckets. Visual feedback is immediate, painted tiles change color, and helps players track their progress. When the pattern is completed correctly, the game ends with a success screen. The wall has a visually worn-out look, adding immersion to the goal of restoration. Background music and subtle sound effects for painting actions maintain the calming atmosphere. The minigame encourages communication, as players must coordinate to efficiently cover the wall without redundant actions. The use of simple movement and interaction controls makes the gameplay accessible and intuitive, while the visual design keeps the experience consistent with the village aesthetic.

ZenGarden is a cooperative 2.5D sokoban-style minigame designed to be a calm yet mentally challenging experience. Players must work together to push pots into their matching target zones - white pots to green targets, light green pots to white targets, and dark green pots to orange targets (it was impossible to find matching colors between pots and targets). The level design emphasizes coordination and careful planning, as there's only one correct sequence of moves to complete the puzzle without getting stuck. Strategic teamwork is essential, especially since placing two white pots simultaneously rewards players with extra points. Despite the difficulty, the minigame maintains a relaxing tone with a soft color palette, soothing background music, and gentle sound effects. The interface includes options to retry or forfeit, supporting a stress-free experience even during trial and error. Upon successful completion, players are rewarded with gardening items, recognizing their effort in mastering what is arguably the most complex minigame in the collection.

Game Engine

This game was developed using Unity. We chose Unity mainly because it makes it easy to implement and customize input controls, such as keyboard and joystick support. Its flexible input system allowed us to quickly configure and test various control schemes, which was essential for the cooperative aspects of the game. Another factor that led us to choose Unity was its ease of use, which makes it what we think is the best “free” game engine available for our purpose.

Concept Art

Some of the 3D models and assets were sourced from Sketchfab. Others were generated using an AI-based pipeline:

1. We used ChatGPT to generate character description prompts because we wanted them to be similar.
2. These prompts were given to Meshy, an AI model, which created the 3D character models.
3. Animations were added using Mixamo, where the generated models were rigged and animated for the game.

Some 3D models, although very few, were also created manually in Unity, making use of the ProBuilder package. Some examples are the logos of the main generator, backup generators, and the house in the Electrical Connections minigame.

Project Plan

Initial Setup and Planning (Week 1–3)

The project began with defining the overall concept and creating the Game Concept Document (GCD), which outlined the story, game mechanics, interface structure, and visual style. Team roles were assigned to distribute responsibilities across design, programming, and art. A simple prototype of the main island was developed to test third-person movement, interaction triggers, and camera control. The Unity project and GitHub repository were initialized with a consistent folder structure to support version control across team members. Simultaneously, an FMOD project was started to organize early audio planning. Basic UI elements such as the main menu, final menu, and navigation between scenes were implemented to facilitate development testing from the start.

Early Development and Asset Preparation (Week 4–6)

The focus shifted toward building early versions of the Electrical Connections, ZenGarden, and Timber Time minigames. The goal was to make their core gameplay systems functional, even with placeholder visuals. Scripts for grid logic, tile rotation, push mechanics, and pot validation were written and tested. The initial version of Timber Time was a completely different minigame. Even though we had all the mechanics implemented, we had to drop it due to a lack of assets that would fit in. Meanwhile, 3D assets were sourced from Sketchfab or generated using AI tools like Meshy and Mixamo, ensuring that all models and animations followed a cohesive art style. Environment assets for the island and each minigame scene were organized and gradually integrated. During this stage, initial versions of the HUD and a

resource tracking system were developed, allowing materials earned from minigames to be stored and visualized in the main game world.

Core Implementation and Testing (Week 7–9)

With basic versions of the first two minigames ready, attention moved to implementing Timber Time and Painting Walls. For Timber Time, the initial version was dropped and the final version created, with the cut-points and axe-cutting mechanics and point system being scripted; for Painting Walls, color matching and brush interaction were coded. Once all minigames had basic functionality, the team began structured internal playtesting to find design flaws, usability issues, and bugs. Tutorials were added to each minigame to guide players and reduce onboarding friction. End screens were created to display results and manage transitions back to the island. Sound and UI feedback were integrated to support game flow and reinforce cooperative actions. Frequent feedback sessions helped improve pacing, level design, and the balance of challenges.

Polishing and Finalization (Week 10–11)

In the final weeks, all minigames were brought to their final state with refined gameplay logic, final art, animations, and sound. The house-building cutscene was scripted, animated, and timed to trigger after all materials were collected. Remaining bugs were resolved, and scene transitions were polished for smooth flow between the island and minigames. FMOD was fully integrated to provide layered sound design, including ambient sounds, character SFX, and UI interactions. Placeholder assets and temporary code were replaced with more quality content. Final visual effects and audio balancing were applied. To support delivery, the team prepared a presentation, updated some of the documentation, and finalized all builds for submission and demonstration.

Test Plan

Testing was divided into two main phases:

- Internal Testing: Each group member tested the parts developed by others, ensuring integration and basic functionality.
- External Testing: We invited people outside the team to play the game. Feedback gathered from this stage led to numerous improvements in gameplay, interface, and overall user experience.

Gameplay Instructions

In the menus:

- Mouse to click buttons

In the main island:

- Player 1: A/D to rotate camera, W/S to move forwards/backwards, ALT to interact, Z to jump

- Player 2: Left/right arrows to rotate camera, up/down to move forwards/backwards, Enter to interact, spacebar to jump
- Mouse to click buttons

In the Timber Time minigame:

- Player 1: A/D to move left/right, E to cut
- Player 2: left/right arrows to move left/right, Enter to cut

In the Electrical Connections minigame:

- Player 1: WASD to move the tile selector and E to make each tile rotate clockwise. Controls the left half of the screen;
- Player 2: up/left/down/right arrows to move the tile selector and Enter to make each tile rotate clockwise. Controls the right half of the screen;

In the Zen Garden minigame:

- Player 1: WASD to move in the directions
- Player 2: Arrows to move in the directions

In the PaitingWalls minigame:

- Player 1: WASD to move in the directions, F to paint and pick colors
- Player 2: Arrows to move in the directions, ENTER to paint and pick colors

Team Members

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